

BUSINESS REQUIREMENTS DOCUMENT FOR KAISER PERMANENTE 1.0

BUSINESS REQUIREMENTS DOCUMENT (BRD)

HealthPulse: AI-Powered Patient Flow & Predictive No-Show Intelligence

(Kaiser Permanente Version 1.0)

1. Document Control

Item	Description
Document Title	Business Requirements Document (BRD) HealthPulse Ai-powered Patient Flow & Predictive No-Show Intelligence
Version	Kaiser Permanente 1.0
Prepared By	Team 10
Date	February 2026
Stakeholders	CEO, CMO, COO, Innovation Lab, Clinic Managers, Data Engineering, IT

2. Executive Summary

HealthPulse Clinic Network has expanded to 15 clinics and 100,000+ patients, creating operational challenges including high no-show rates, long wait times, uneven provider utilization, and fragmented data systems.

This project introduces an AI-powered predictive no-show model, a centralized data warehouse, and an operational intelligence dashboard to enable proactive decision-making, reduce revenue leakage, and improve patient experience.

The initiative is expected to deliver \$3.2M net benefit over 5 years with a one-time investment of \$250K.

3. Business Objectives

- Reduce no-show rate below 8%
- Reduce average patient wait times to under 10 minutes
- Balance provider utilization to 70–85%
- Centralize operational data into a unified analytics layer
- Enable predictive, proactive operational management

4. Problem Statement

HealthPulse currently experiences:

- No-show rates above 12%
- \$1.2M annual revenue loss from empty appointment slots
- Patient wait times exceeding 15 minutes
- Provider utilization imbalance (40%–95%)
- Fragmented data across multiple systems
- No predictive capability to anticipate operational issues

These issues hinder scalability, efficiency, and patient satisfaction.

5. Scope

In Scope

- Predictive no-show model
- Data warehouse (star schema)
- Real-time prediction API
- Operational dashboard
- Documentation (model card, API spec, data dictionary)

Out of Scope

- EHR replacement

- Patient mobile app
- Automated scheduling optimization engine (future phase)

6. Stakeholder Analysis

Stakeholder	Role	Interest
CEO	Executive Sponsor	ROI, competitive advantage
CMO	Clinical Oversight	Provider satisfaction, patient outcomes
COO	Operations Lead	Efficiency, throughput
Clinic Managers	Operational Users	Scheduling efficiency
Data Engineering	Builders	Pipelines, warehouse
Innovation Lab	Owners	AI model development
Scheduling Team	Daily Users	No-show risk visibility
Patients	Indirect Users	Reduced wait times

7. Business Requirements

BR-1: Predict high-risk no-show appointments with $\geq 75\%$ accuracy

BR-2: Trigger automated reminders for high-risk patients

BR-3: Provide real-time dashboards for utilization and no-show risk

BR-4: Consolidate operational data into a unified warehouse

BR-5: Deliver real-time API responses in $< 300\text{ms}$

BR-6: Ensure scalability to 30+ clinics

8. Functional Requirements

8.1 Predictive Model

- FR-1: Accept historical appointment, patient, and provider data
- FR-2: Generate risk score (0–1) and category (Low/Medium/High)
- FR-3: Provide top contributing factors (explainability)
- FR-4: Support retraining with new data

8.2 Data Warehouse

- FR-5: Ingest data from EHR, scheduling, and patient systems
- FR-6: Maintain fact tables (Appointments, Providers, Clinics)
- FR-7: Maintain dimension tables (Date, Time, Patient, Provider)
- FR-8: Run daily ETL with validation checks

8.3 Prediction API

- FR-9: Accept JSON payload with appointment attributes
- FR-10: Return risk score in <300ms
- FR-11: Log all requests and responses
- FR-12: Require token-based authentication

8.4 Dashboard

- FR-13: Display no-show risk by clinic, provider, date
- FR-14: Show provider utilization trends
- FR-15: Allow drill-down to appointment level
- FR-16: Export reports to CSV

9. Non-Functional Requirements

Performance

- NFR-1: API latency <300ms
- NFR-2: Dashboard loads <5 seconds

Security

- NFR-3: HIPAA-compliant data handling
- NFR-4: Encryption in transit and at rest

Scalability

- NFR-5: Support 10,000+ predictions/day
- NFR-6: Expandable to 30+ clinics

Reliability

- NFR-7: 99.5% API uptime
- NFR-8: Automated pipeline monitoring

Usability

- NFR-9: Dashboard intuitive for non-technical staff
- NFR-10: Mobile-friendly UI

10. AS-IS vs TO-BE Process Flows

AS-IS

1. Appointment booked
2. Scheduler manually confirms
3. No visibility into no-show risk
4. Patients forget or skip appointments
5. Providers experience uneven workloads
6. Managers react after issues occur
7. Data scattered across systems

TO-BE

1. Appointment booked
2. Prediction API scores risk instantly
3. High-risk patients receive automated reminders
4. Dashboard shows real-time utilization and risk
5. Managers proactively adjust schedules
6. Providers have balanced workloads
7. Unified data warehouse supports analytics

11. KPIs & Success Metrics

Category	KPI	Target
Operational	No-show rate	
Operational	Avg wait time	

Operational	Provider utilization variance	70–85%
Financial	Revenue recovered	\$380K/year
Financial	Utilization improvement	\$200K/year
Technical	Model accuracy	≥75%
Technical	API latency	

12. Assumptions

- Historical appointment data is available
- Scheduling system can integrate with API
- Staff will adopt new workflows with training
- Data quality is sufficient for modeling

13. Risks & Mitigations

Risk	Impact	Mitigation
Poor data quality	Low model accuracy	Data cleaning, validation rules
Staff resistance	Low adoption	Training, change management
Model bias	Patient inequity	Bias testing, fairness checks
API downtime	Operational disruption	Redundancy, monitoring

14. Success Validation Plan

- Compare no-show rate pre- vs post-deployment

- Validate model accuracy on holdout data
- Conduct staff usability testing
- Measure API performance under load
- Validate data completeness in warehouse

15. Approval Sign-Off

Name	Role	Signature	Date
CEO	Executive Sponsor		
CMO	Clinical Lead		
COO	Operations Lead		
Head of Data	Technical Lead		
Project Manager	Delivery Lead		